

XINYU MAO

Email: xinyumao.tcs@gmail.com

Website: sparkmxy.github.io

RESEARCH INTEREST

I am broadly interested in theoretical computer science, especially in cryptography and computational complexity.

EDUCATION

University of Southern California Ph.D. student in Computer Science	Aug. 2022 – Present Los Angeles, United States
Shanghai Jiao Tong University Bachelor of Engineering in Computer Science	Aug. 2018 – June 2022 Shanghai, China

RESEARCH EXPERIENCE

University of Southern California Research Intern advised by Prof. Jiapeng Zhang Research Topic: foundations of cryptography.	May 2021 - April 2022
Shanghai Jiao Tong University Undergraduate Researcher advised By Prof. Shengli Liu Research Topic: public-key cryptography	Sept. 2020 – Feb. 2021

PUBLICATIONS

1. Xinyu Mao, Noam Mazon, and Jiapeng Zhang. Non-adaptive universal one-way hash functions from arbitrary one-way functions. In *Advances in Cryptology – EUROCRYPT 2023: 42nd Annual International Conference on the Theory and Applications of Cryptographic Techniques, Lyon, France, April 23-27, 2023, Proceedings, Part IV*, page 502–531, Berlin, Heidelberg, 2023. Springer-Verlag
2. Mi-Ying Huang, Xinyu Mao, Guangxu Yang, and Jiapeng Zhang. Communication lower bounds of key-agreement protocols via density increment arguments. In Guy Rothblum and Hoeteck Wee, editors, *Theory of Cryptography*, pages 185–206, Cham, 2023. Springer Nature Switzerland
3. Xinyu Mao and Jiapeng Zhang. On the power of svd in the stochastic block model. *Advances in Neural Information Processing Systems*, 36:32765–32781, 2023
4. Xinyu Mao, Guangxu Yang, and Jiapeng Zhang. Gadgetless lifting beats round elimination: Improved lower bounds for pointer chasing. *Innovations in Theoretical Computer Science (ITCS)*, 2025
5. Yilei Chen and Xinyu Mao. Universal computational extractors and multi-bit aipo from lattice assumptions. In Serge Fehr and Pierre-Alain Fouque, editors, *Advances in Cryptology – EUROCRYPT 2025*, pages 63–93, Cham, 2025. Springer Nature Switzerland
6. Mi-Ying Miryam Huang, Xinyu Mao, Shuo Wang, Guangxu Yang, and Jiapeng Zhang. A min-entropy approach to multi-party communication lower bounds. In *40th Computational Complexity Conference (CCC 2025)*, pages 33–1. Schloss Dagstuhl–Leibniz-Zentrum für Informatik, 2025

HONORS AND AWARDS

Zhiyuan Outstanding Student Scholarship	2022
Shanghai Outstanding Graduate	2022

Foresight-Sequoia Scholarship 70 winners in SJTU each year	2021
Zhiyuan Honorary Scholarship Top 5% in SJTU each year	2018, 2019, 2020
Bronze Medal The 34th China National Olympiad in Informatics (NOI)	2017

TEACHING EXPERIENCE

CSCI 575: Quantum Computing and Quantum Cryptography Teaching Assistant	Fall 2025 University of Southern California
CSCI 531: Applied Cryptography Teaching Assistant	Spring 2025 University of Southern California
CSCI 270: Introduction to the Theory of Computation Teaching Assistant	Fall 2023 University of Southern California
Great Ideas in Computer Science Teaching Assistant I wrote an <u>Enigma Simulator</u> to facilitate teaching.	Fall 2019, Fall 2020, Fall 2021 Shanghai Jiao Tong University

SKILLS

Programming: C++, Java, Python, JavaScript, Latex
Language: Chinese (native), English (TOEFL 105)